SUPPORTING ADULT GUIDANCE

Surds: meaning and simplifying

Number Systems and Calculator Judgement · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson introduces surds and simplifying. Learners see that no rounded decimal squares to exactly 2, sort roots into exact values and surds, and test whether $\sqrt{a} \blacksquare \sqrt{b} = \sqrt{(a \blacksquare b)}$ for each operation. In the main activity, dots regroup into square blocks to show $\sqrt{72} = 6\sqrt{2}$, and learners find the largest square factor. They then add like surds and reverse the process. Two optional proofs are included: $\sqrt{2}$ is irrational, and $\sqrt{a} \times \sqrt{b} = \sqrt{(ab)}$.

Driving Question: *Why keep √2 instead of writing 1.414…?*

The mathematics, in plain terms

A surd is an irrational root, such as $\sqrt{2}$ or $\sqrt{12}$. For $a, b \geq 0$, $\sqrt{(ab)} = \sqrt{a} \times \sqrt{b}$, because both sides are non-negative and square to ab. So $\sqrt{(k^2 m)} = k\sqrt{m}$; a surd is in simplest form when the number under the root has no square factor other than 1. There is no corresponding rule for addition: $(\sqrt{a} + \sqrt{b})^2 = a + b + 2\sqrt{(ab)}$. The classic proof that $\sqrt{2}$ is irrational assumes $\sqrt{2} = p/q$ in lowest terms and shows both p and q must be even.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The square-factor finder

Simplify $\sqrt{72}$.

1. Try 4: $72 = 4 \times 18$, so $\sqrt{72} = 2\sqrt{18}$.	true, but $18 = 9 \times 2$ still has a square factor
2. Find the largest square factor of 72.	$72 = 36 \times 2$
3. 72 dots make 2 blocks of 6×6.	$\sqrt{72} = \sqrt{36} \times \sqrt{2}$
4. Take the root of 36 outside.	$\sqrt{72} = 6\sqrt{2}$
5. Check: $6^2 \times 2 = 36 \times 2 = 72$.	✓

$\sqrt{72} = 6\sqrt{2}$. The largest square factor, 36, does it in one step.

Example 2 — Adding surds

Work out √12 + √75.

1. Simplify each surd first.	√12 = √4 × √3 = 2√3
2. Then the second one.	√75 = √25 × √3 = 5√3
3. Both are multiples of √3: like surds.	2√3 + 5√3
4. Collect them, like 2x + 5x.	= 7√3

√12 + √75 = 7√3. (√12 + √75 is not √87.)

Pattern, then proof

Learners first see that rounding never reaches exactness, then test the multiplication rule on square numbers (where they can check it), and only then use it to simplify. The dot blocks make 'largest square factor' visible: 72 dots regroup as 2 blocks of 36. The proof stretches justify the rule and answer the Driving Question — √2 has no exact decimal, so we keep the symbol. Encourage ready learners to try them; the secure core is simplifying with the largest square factor. No calculators.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Exact or rounded? Square rounded √2 values; sort exact values and surds. Stretch: prove √2 is irrational.	~8 min
Movement	Does the rule hold? Test √a ■ √b against √(a ■ b) for each operation.	~9 min
Reflection	The square-factor finder: dots regroup into square blocks. Stretch: prove √a × √b = √(ab).	~15 min
Creativity	Surd pairs: find every pair of tiles that adds to 7√3; then back under one root.	~8 min
Rest	A length you can draw perfectly but never write exactly as a decimal.	~2 min
Nutrition	Intellectual: an appreciation of exactness.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

$\sqrt{9} + \sqrt{16} = \sqrt{25}$ (adding under the root).
Ask: “What are $\sqrt{9}$ and $\sqrt{16}$? What is their sum? What is $\sqrt{25}$?”

Simplifying with a smaller square factor and stopping ($\sqrt{72} = 2\sqrt{18}$).
Ask: “Does 18 have a square factor too?”

Multiplying the outside number in, without squaring it ($3\sqrt{5} = \sqrt{15}$).
Ask: “3 is the square root of what?”

Adding unlike surds ($\sqrt{2} + \sqrt{3} = \sqrt{5}$).
Ask: “Are they the same kind of surd, like $2x$ and $5x$?”

Thinking 1.414 is the exact value of $\sqrt{2}$.
Ask: “What is 1.414 × 1.414?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: square numbers and factors. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes $\sqrt{72} = 2\sqrt{18}$ and stops. The aim is a question that lets them find the next step themselves.

- “Is $2\sqrt{18}$ equal to $\sqrt{72}$? How could you check?”
- “Does 18 have a square factor?”

- “Which square number was hiding in 72 all along?”
- “Can you get there in one step?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Are the proofs required?
No — they are stretches. The secure core is recognising surds and simplifying with the largest square factor.

Is $2\sqrt{18}$ wrong?
It is equal to $\sqrt{72}$, so the value is right, but it is not fully simplified. The lesson recognises the value and asks the learner to simplify further.

Why no calculators?
A calculator gives a rounded decimal, which is exactly what surds avoid.

What comes next?
Lesson 6: manipulating surds and rationalising the denominator.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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